

Rhetorical Cohesion: Measuring Party Unity in Political Messaging

Gechun Lin*

Benjamin S. Noble[†]

Abstract

Party messaging is central to the study of elections, legislative behavior, and democratic accountability. Yet, we lack a scalable measure of whether party messaging is cohesive. Existing text measures capture specific features of that messaging, like ideology or partisanship, but not similarity of the overall message. To address this gap, we develop a framework for measuring rhetorical cohesion—the degree to which co-partisans say the same things about the same issues at the same time. We leverage topic models and contextual text embeddings to construct our measure, which we validate against human judgment. We show it exhibits theoretically-expected relationships with electoral timing, majority control, issue salience, and seat safety, but when we substitute plausible feature-specific alternatives, coefficients weaken or reverse. Our framework allows scholars to study the rhetorical side of party unity and better understand how party brands are built.

*Assistant Professor, Department of Political Science, Texas A&M University. [lingechn.github.io](https://github.com/lingechn). lingechn@tamu.edu.

[†]Assistant Professor, Department of Political Science, University of California, San Diego. benjaminoble.org. b2noble@ucsd.edu.

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The authors used ChatGPT and Claude (models varied across the period of use) in 2026 to assist with manuscript preparation. Specifically, these models were used to suggest copyedits, provide feedback on organization, aid in the creation of Figure 1, and offer simulated peer review on manuscript drafts. All suggestions were reviewed and potentially revised by the authors. ChatGPT and Claude (models varied across the period of use) were also used during 2025 to 2026 as a coding assistant in the implementation of the computational analyses. Models were used to help draft portions of analysis code and to identify and debug programming errors. The authors reviewed, modified, and executed the resulting code and verified the analytical outputs.

1 Introduction

Individual politicians seek re-election (Mayhew 1974), but their electoral fortunes are tied to the party brand, the “actions, beliefs, and outcomes commonly attributed to the party as a whole” (Cox and McCubbins 1993, 10). These party brands are collective goods shaped by the actions of all members (Aldrich 1995), and they are strengthened when individuals present a cohesive public image (Greene and Haber 2015; Groeling 2010). Lawmakers shape this image not only through their votes but also through their voices. When co-partisans promote a consistent message, it is more likely to reach, and stick with, an inattentive public (Groeling 2010; Sellers 2009). Party leaders act accordingly: they develop messaging themes, distribute talking points, and spend considerable effort recruiting the rank-and-file to echo them (Evans 2001; Evans and Oleszek 2001; Harris 2005; Lee 2016). Yet members may depart from a common message to appeal to distinct constituencies, and parties may deliberately diversify their appeals across audiences. How, then, can we measure the extent to which party members speak with a common voice?

We call this construct *rhetorical cohesion*: the extent to which members of a party align in their communication about a given policy issue. Measuring rhetorical cohesion is challenging because it is a latent construct that can manifest in many ways. Researchers observe individual messages and their associated attributes, but not cohesion itself. Messages may share distinctive language (Gaynor 2025; Gentzkow, Shapiro and Taddy 2019), express a common political position (Lauderdale and Herzog 2016; Gaynor et al. 2025), deploy similar arguments (Blumenau and Lauderdale 2024), or employ the same rhetorical appeals (Ban and Kaslovsky 2025; Müller and Proksch 2024). Existing text-as-data methods provide theoretically grounded measures of these specific forms of alignment, but no single feature is equivalent to rhetorical cohesion. Party members can align along one dimension while conveying quite different messages. Measuring rhetorical cohesion therefore requires targeting overall message similarity.

To address this challenge, we introduce a general framework for measuring rhetorical

cohesion at scale. The framework first identifies messages on a common substantive issue within a defined time period and then uses contextual embeddings to estimate their overall similarity. We apply this framework to House floor speeches from 1961–2020 and subject our measures of rhetorical cohesion to several validation tests. These measures closely recover human judgments of overall rhetorical similarity and outperform existing measures based on specific rhetorical features. Our rhetorical cohesion scores are positively associated with related forms of alignment, including language reuse and expressed political positions, without being reducible to them. They also recover a well-documented historical case: non-Southern Democrats are more rhetorically cohesive than cross-factional pairs combining one Southern and one non-Southern Democrat from 1961–1994. This gap attenuates after. We then use our measures to examine the political correlates of rhetorical cohesion. Cohesion is higher among majority-party members and electorally secure lawmakers, on more salient issues, and as midterm elections approach. Importantly, substituting feature-specific text measures for rhetorical cohesion weakens or even reverses these relationships.

The contribution is methodological as well as substantive. We provide a validated framework for converting unstructured political text into a measure of collective rhetorical alignment without requiring researchers to specify, in advance, the particular linguistic or substantive features on which actors must agree. By producing a systematic and scalable measure, our framework enables researchers to study the strategic variation of rhetorical cohesion across political contexts. Although we focus on party cohesion in the U.S. Congress, the framework can be adapted to comparative legislative settings, enabling cross-national measurement of rhetorical cohesion on a common scale when paired with consistent topic classification schemes. Moreover, it opens new avenues for examining the sources and consequences of rhetorical cohesion. Researchers can investigate how increasingly prominent dimensions of political communication—including emotional appeals, partisan language, and moral rhetoric—shape rhetorical cohesion, and, in turn,

how that cohesion influences party brands, voter perceptions, and democratic accountability.

First, we conceptualize rhetorical cohesion and discuss the challenges of measuring this latent construct. We then introduce our two-stage measurement framework and demonstrate how to apply it to a large corpus of congressional speeches. Next, we validate the resulting measure against human judgment, feature-specific measures of rhetorical alignment, and a well-documented historical pattern of party cohesion. Finally, we demonstrate its substantive application by examining how rhetorical cohesion varies under different political conditions: our measure yields patterns consistent with theoretical expectations while feature-specific measures do not always fully recover these same relationships. We conclude with a discussion of the limitations and potential of our method.

2 Forms of Rhetorical Alignment

2.1 Conceptualizing Rhetorical Cohesion

The legislative politics literature defines party cohesion as “the extent to which, in a given situation, group members can be observed to work together for the group’s goal in one and the same way” (quoted in Hazan 2003, 3), particularly with respect to policy agreement (Sinclair 2003). We extend this idea from legislative behavior to party communication, conceptualizing rhetorical cohesion as the degree to which members of a party communicate cohesively about a given policy issue.

To operationalize this concept, we draw on recent work by Mueller (2024), which measures crowd cohesion through the degree of semantic agreement in protesters’ expressed demands and shows that protests with more cohesive demands are more likely to secure concessions aligned with those demands. Similarly, we operationalize rhetorical cohesion as the overall similarity of messages conveyed by party members on the same policy issue, rather than as agreement along any single observable dimension. Such similar-

ity may arise from a combination of substantive agreement (e.g., shared issue positions or ideology) and linguistic coordination (e.g., similar wording or repeated phrases), but cannot be reduced to any one of these dimensions.

2.2 Challenge of Measuring Rhetorical Cohesion

Measuring rhetorical cohesion poses a distinct methodological challenge. Political scientists regularly use text to estimate difficult-to-observe quantities, but these methods are typically designed to recover a particular feature of communication rather than whole-message similarity. For example, text-scaling methods place speakers on a latent dimension, often interpreted as a rhetorical ideal point (Laver, Benoit and Garry 2003; Slapin and Proksch 2008; Lauderdale and Herzog 2016; Gaynor et al. 2025). More recently, scholars have turned to classification methods to measure rhetorical partisanship and polarization (Gentzkow, Shapiro and Taddy 2019; Noble and Butler 2026; Peterson and Spirling 2018). These approaches, which we refer to as *expressed ideology* and *rhetorical partisanship*, are valuable constructs, but neither is equivalent to rhetorical cohesion. A text-scaling method may place similar speeches farther apart if they express that similarity using different words. A classification model may assign two speeches to the same party even when they communicate different messages. More generally, any measure built around a single rhetorical feature risks conflating one aspect of political communication with the overall consistency of party messages.

Lexical overlap measures offer a way to more directly quantify message similarity (Casas, Denny and Wilkerson 2020; Gaynor 2025). However, these are ambiguous indicators of rhetorical cohesion. Speeches with little shared language may still communicate the same message using different words. Speeches that share a set of distinctive talking points may still make different arguments. Table 1 illustrates this ambiguity and the broader challenge by excerpting three Republican floor speeches on H.R. 3962, the Affordable Health Care for America Act, delivered on November 3, 2009. A standard 5-gram

Table 1: Different Forms of Rhetorical Alignment in Three Health Care Speeches

Panel A. H.R. 3962 Speech Excerpts

- Q1 Rep. Joseph Pitts (R-PA):**
 “The Pelosi health care bill we will consider later this week effectively eliminates the popular Medicare Advantage health plans that millions of seniors rely on for medical, vision, and dental care.”
- Q2 Rep. John Fleming (R-LA):**
 “The CBO estimates in PelosiCare that it will cut over \$150 billion from Medicare Advantage...That will knock about 6 to 11 million seniors off of Medicare Advantage.”
- Q3 Rep. Mike Pence (R-IN):**
 “The American people want health care reform that lowers the cost of health insurance rather than increasing the cost of government. But it looks like that’s exactly what they’re getting in the Pelosi health care bill.”

Panel B. Illustrative Forms of Alignment

Selected Feature	Alignment	Why
Lexical overlap	Q1 and Q3	Both use “the Pelosi health care bill.”
Affected groups	Q1 and Q2	Both discuss seniors.
Cost-benefit framing	Q2 and Q3	Both emphasize financial consequences.
Issue attention	All three	All discuss health care.
Stance	All three	All oppose H.R. 3962.
Partisan framing	All three	All reference Speaker Pelosi by name.

Note: Panel A presents three excerpts from speeches on HR 3962 on 11/3/2009. Panel B illustrates how these speeches can differentially align on specific features.

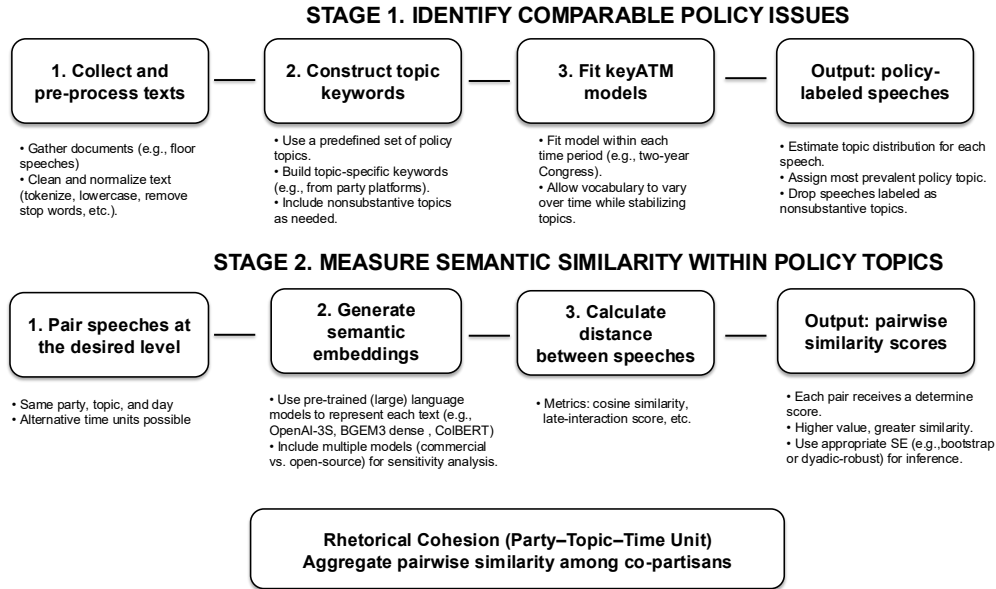
overlap measure would identify Q1 and Q3 as the most similar pair given their shared use of the phrase “the Pelosi health care bill.” However, comparisons based on other features produce different rankings. For example, attention to groups would highlight Q1 and Q2, given their focus on seniors. A cost-benefit emphasis would rank Q2 and Q3 highest. Measures of topic-based issue attention, stance, and partisan framing might rank all three speeches as equally similar. There is no “right” measure other than the one suggested by theory, and each captures a genuine similarity or difference between speech-pairs. However, measuring rhetorical cohesion requires that we target *overall* message similarity, not just one feature.

A newer class of semantic similarity methods based on text embeddings offers a different approach to comparing political texts (Rheault and Cochrane 2020; Lin 2025; Ornstein 2026; Alvarez and Morrier 2026). Rather than defining similarity in terms of pre-specified features, these methods represent texts as numeric vectors in a high-dimensional semantic space learned from statistical patterns in how words and phrases co-occur in natural language. As a result, speeches expressing similar political messages can receive nearby vector representations even when they differ in word choice and syntax (Rodriguez and Spirling 2022; Rodriguez, Spirling and Stewart 2023). This property makes text embeddings well suited for operationalizing rhetorical cohesion. The tradeoff is that embedding models do not identify *which* rhetorical features give rise to a similarity score, nor can we assume that model-derived similarities correspond to how humans perceive rhetorical similarity. Consequently, we propose our method for cases where scholars are interested in whole-message similarity, and we recommend existing measures when a specific feature is of interest. We also empirically establish a strong correspondence between our method’s judgments of overall message similarity and human interpretation.

3 A General Framework of Measuring Rhetorical Cohesion

We construct our measure of rhetorical cohesion in two stages. First, we identify documents that cover comparable policy issues using customized topic modeling. This step allows us to pair documents that discuss the same policy, which is essential for operationalizing our definition of rhetorical cohesion. Second, we capture message variation within issues by measuring the semantic similarity between co-partisans’ same-issue statements using contextual text embeddings. These two stages provide a fine-grained measure of rhetorical policy cohesion, i.e., speaking “in one way about the same issue.” Figure 1 summarizes the two-stage workflow we use to construct our measure of rhetorical cohesion. More details are provided as follows.

Figure 1: Workflow of Measuring Rhetorical Cohesion.



Note: GPT-5.6 Sol (August 2026) was used to generate an initial version of the workflow figure. The authors subsequently revised the content and created the final figure.

3.1 Identifying Topics: Comparable Policy Issues Across Time

We demonstrate our approach with an application to U.S. House floor speeches. Although our two-stage workflow is agnostic to the corpus, we choose House floor speeches to mirror other recent text-as-data work quantifying rhetorical similarities and differences among lawmakers (Gaynor et al. 2025; Gentzkow, Shapiro and Taddy 2019; Rodriguez and Spirling 2022).

We begin by identifying the most prevalent topic of each speech using a keyword assisted topic model (keyATM), a semi-supervised approach that allows the researcher to specify topics with relevant keywords before fitting (Eshima, Imai and Sasaki 2024). This model enables us to focus on a set of broad, stable policy issues that we expect to persist across our time series, facilitating over-time comparisons. Our topics come from the Comparative Agendas Project (CAP), which categorizes political content into a set of

21 major topics such as macroeconomy, health, and foreign affairs (Jones et al. 2023). To generate keywords, we downloaded the CAP version of the Democratic and Republican Party Platforms, which have been hand-coded at the quasi-sentence level by Wollbrecht et al. (2023). We calculated the tf-idf score of each stemmed word in this corpus within each topic and chose the top 15 words in each category as our topic-specific keywords (see Table A1 for the full list). We manually created two additional non-CAP topics specific to congressional speeches: *parliamentary language* (e.g., quorum, yield) and *other*, which included uninformative and common words (e.g., people, think). We apply standard pre-processing to all speech transcripts (see Grimmer, Roberts and Stewart 2022, and Appendix A.2) and fit a separate keyATM model to all speeches delivered within each two-year Congress. This approach allows vocabulary and policy content to vary over time, while the use of keyATM stabilizes topics.

After fitting, congressional speeches are characterized by mixed-membership topic distributions. However, we assign each speech a single label according to its most prevalent topic and drop speeches labeled *parliamentary* or *other*, because our measure of rhetorical cohesion is defined as the similarity of speeches addressing a common policy issue.¹ Following Ying, Montgomery and Stewart (2022), we validate that speeches are thematically related to their assigned topics in Appendix A.3.

3.2 Beyond Topics: Capturing Party Cohesion with Semantic Similarity

Topics provide a useful way to identify the policy issue a speech addresses, but they are inherently coarse. As Waight et al. (2025) argue, topic models group documents by broad themes, but they do not distinguish the specific claims, arguments, or framing they contain. This limitation is particularly important for congressional floor speeches, whose

¹While topic-probability weighting schemes are possible, hard-assignment provides an interpretable operationalization while making the analysis computationally feasible.

topics are often shaped by procedural rules and party leaders’ agenda-setting power. We therefore move beyond topic-level analysis by representing each speech with a dense semantic embedding. Unlike Waight et al. (2025), who use SBERT to retrieve candidate document pairs and then apply LLM-based claim extraction and pairwise annotation to determine whether two news articles express the same underlying narrative, our objective is to quantify the degree of rhetorical similarity between speeches rather than train a binary narrative classifier.

We employ OpenAI’s commercial contextual embedding model, *text-embedding-3-small* (hereafter OpenAI-3S), alongside BGEM3, an open-source embedding model trained for semantic retrieval using both dense and ColBERT representations.² These embedding models offer three advantages. First, both OpenAI and BGEM3 embeddings leverage transformer architectures that learn word meanings in context through language-modeling or contrastive retrieval objectives. As a result, texts that express similar ideas, even when using different words, are represented as “close” in the embedding space. These models capture semantic relatedness rather than exact word overlap. Second, these models produce fully contextualized embeddings—token representations vary with their usage, which is important in political contexts. They improve upon static word embeddings such as word2vec (Mikolov et al. 2013) and GloVe (Pennington, Socher and Manning 2014), which assign fixed vectors to words regardless of context (see Lin 2025, for a comprehensive discussion). Third, advances in model scale and architecture (Minaee et al. 2024) allow these embedding models to handle substantially longer input sequences than earlier pre-trained language models. Existing research also demonstrates that LLMs and their derived embeddings generalize well in zero- and few-shot settings (Rathje et al. 2024; Tao et al. 2024), whereas modern retrieval-trained embeddings such as BGEM3 pro-

²BGEM3 is a transformer-based retrieval model that produces both dense embeddings, which represent documents as single vectors optimized for semantic similarity, and ColBERT-style embeddings, which encode texts as sets of contextualized token-level vectors enabling fine-grained, interaction-based similarity matching (Chen et al. 2024). OpenAI-3S models are derived from LLMs in the GPT family and learn contextual semantic representations from large-scale corpora using language-modeling-based objectives. Full details are not publicly disclosed (Yenduri et al. 2024).

vide strong off-the-shelf performance for semantic similarity, making both scalable for large, unlabeled political text corpora.

To compute our measure, we compare speeches at the day-party-topic level: we pair each speech given by a lawmaker on a topic with each other speech given by a co-partisan member on the same topic on the same day.³ By restricting pairs to the same period of time, we maximize the likelihood that speeches on the same topic cover the same underlying sub-issue. However, researchers can choose their unit of analysis as befits their study. Then, we calculate a similarity score for each pair of texts. For OpenAI-3S and BGEM3-dense models, we use cosine similarity, following OpenAI’s official guide⁴ and the BGEM3 model documentation on Hugging Face.⁵ Cosine similarity is theoretically bounded between -1 and 1 , with higher values indicating greater semantic similarity. In our data, all cosine similarities fall between 0 and 1 , reflecting both the properties of these embedding models and that speeches are paired within the same topic. The BGEM3-ColBERT model uses a late-interaction mechanism that matches each token in one statement with its most similar token in the other and averages these token-level similarities to produce a normalized semantic alignment score between 0 and 1 . After selecting an embedding model (and conditioning on the text and its assigned topic), the pairwise scores are deterministic. As such, we cannot report standard errors or confidence intervals on individual scores. Instead, we report bootstrap and dyadic-robust standard errors where appropriate and replicate our results across embedding models to assess the sensitivity of our results.

The resulting similarity score is a scalar summary of rhetorical cohesion: it identifies which pairs of speeches are more or less rhetorically cohesive without imposing a left-right (or any other low-dimensional) interpretation on the data. We can use these pair-

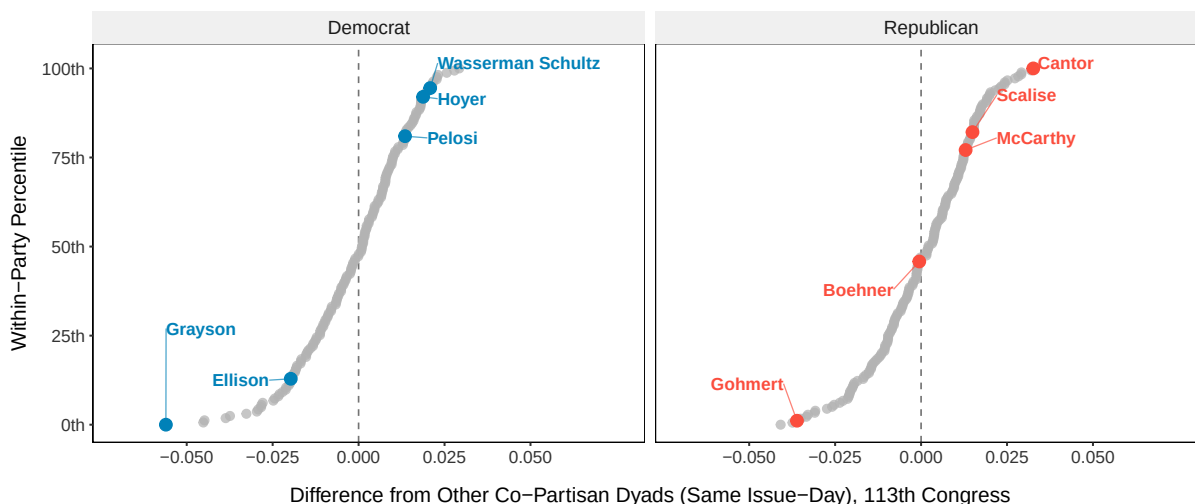
³We exclude independents and party-switchers in the Congress they switch. Covariates come from VoteView and The Center for Effective Lawmaking). We exclude a small subset of lawmakers for whom we do not have key covariates.

⁴See <https://developers.openai.com/api/docs/guides/embeddings>.

⁵See <https://huggingface.co/BAAI/bge-m3>.

wise scores to describe individual-level rhetorical alignment, for example, at the two-year Congress level. Here, we average all pairwise ColBERT scores within each member-pair, topic, and day. Then, for each topic-day, we average across all of a member’s distinct co-partisan counterparts. We compare this value to the mean cohesion of all co-partisan pairs discussing the same topic on that day (excluding that member) and average across all topic-days in that Congress.⁶ Positive (negative) scores indicate a member’s rhetoric is more (less) aligned with her party than a typical co-partisan.

Figure 2: Individual Rhetorical Alignment, 113th Congress



Note: Each point represents a House member. The x -axis denotes a member’s average rhetorical similarity relative to the mean similarity across all co-partisan dyads (excluding that member) discussing the same topic on the same day. Positive (negative) values indicate more (less) alignment. The y -axis ranks members by within-party percentile.

We plot these scores and their within-party percentile ranks for the 113th House in Figure 2. Here, we observe facially valid estimates for key lawmakers. Then-DNC Chair, Wasserman Schultz and Democratic Leaders Pelosi and Hoyer appear near the top of the distribution. Strong progressives, Ellison and Grayson, appear near the bottom. On the Republican side, Cantor, McCarthy, and Scalise, members of Republican leadership, appear near the top. Strong conservative Gohmert appears near the tail. Interestingly,

⁶We drop all member-topic-days on which a lawmaker has only one co-partisan counterpart. Then, we drop members with fewer than 10 remaining topic-days.

Boehner is slightly less rhetorically cohesive than the median Republican despite being the Speaker. This ranking is consistent with the intra-party conflict that plagued his speakership and led to his resignation.

4 Validating our Measures

We conceptualize rhetorical cohesion as a latent construct manifested in the overall similarity of party members’ messages, so its validity cannot be established by interpreting individual embedding dimensions. Instead, we empirically evaluate whether the resulting similarity scores match human judgments of rhetorical similarity, exhibit the expected relationships with specific rhetorical features, and recover well-documented patterns of party cohesion.

4.1 Construct Validity Against Human Judgment

To determine whether our measure of rhetorical cohesion accords with human judgments of speech similarity, we worked with a trained undergraduate RA and a set of crowd-workers to conduct a novel pairwise validation exercise (cf. Carlson and Montgomery 2017). We presented each rater with a *focal speech* sampled from our corpus between 1994–2016 along with two *comparison speeches* that were paired with the focal speech in the data. The raters were then asked to complete a series of *comparison tasks*. For each, they carefully read the focal speech and both comparison speeches, then they selected the comparison speech that was more similar to the focal speech, where similarity was broadly defined. We sampled several comparison tasks for each focal speech for a total of 249 tasks. Each task was completed by our RA and two crowd workers. All three coders selected the same comparison speech in 63% of tasks. Agreement was lower when two speeches had a smaller gap on the embedding-based measures, indicative of more difficult comparisons. We discuss this validation task and present additional inter-coder

reliability statistics in Appendix A.4.

Our quantity of interest is the proportion of times a given measure selects the comparison speech chosen by majority vote among the human raters. The similarity scores derived from three contextual embedding models—OpenAI-3S, BGEM3-Dense, and BGEM3-ColBERT—matched in 83.9%, 84.3%, and 87.6% of tasks, respectively. We also compare the performance of our measure to three popular text-alignment measures that, while designed for other purposes, could potentially serve as alternatives. We define these measures here and use them in the remainder of our validation and empirical exercises:

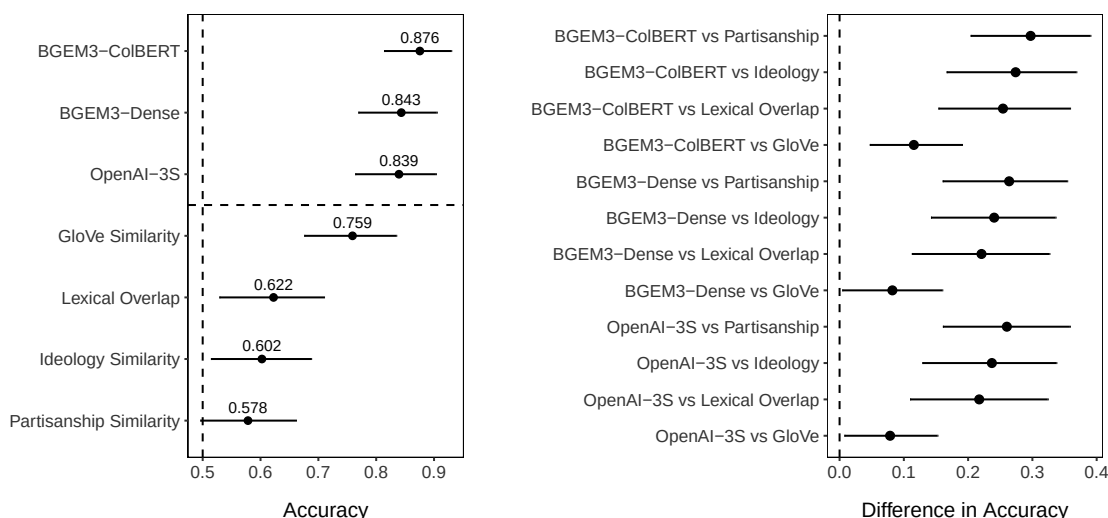
- **Expressed Ideology:** We compute Wordfish scores for all speeches within each Congress-topic subset and take the negative absolute difference between each pair of speeches (cf. Gaynor et al. 2025; Lauderdale and Herzog 2016; Slapin and Proksch 2008).
- **Rhetorical Partisanship:** We train five-fold ridge classifier to predict whether an out-of-fold speech was delivered by a Republican within each Congress subset. We compute the negative absolute difference between these probabilities for each pair of speeches (cf. Gentzkow, Shapiro and Taddy 2019; Noble and Butler 2026; Peterson and Spirling 2018).
- **Lexical Overlap:** We divide each speech into shingled 5-grams and compute set Jaccard similarity speech pairs (cf. Gaynor 2025; Casas, Denny and Wilkerson 2020).

For the human validation benchmark, we also compare our measures to a static embedding measure, GloVe (e.g., Rodriguez and Spirling 2022). We average each word’s GloVe embedding within a speech to produce a single speech vector. We compute cosine similarity between speech vectors. All measures are specified so higher values indicate higher text alignment. These match the human raters when they assign a higher value to the selected focal-comparison pair.

In the left panel of Figure 3, we show accuracy and bootstrapped confidence intervals for our three measures and the four alternatives. Our measures of rhetorical cohesion more accurately recover the selections made by the majority vote of the human raters, with BGEM3-ColBERT performing best. Lexical overlap, expressed ideology, and partisan similarity, which capture specific aspects of alignment and are not expected to reproduce rhetorical cohesion, perform significantly worse at recovering human judgment of overall speech similarity, with accuracies of 0.622, 0.602, and 0.578 respectively.⁷ GloVe performs better than these feature-specific measures (accuracy 0.759), but falls short of our measures. As shown in the right panel, our measures significantly outperform in recovering human judgments of overall speech similarity.

Figure 3: Accuracy of Embedding Methods vs Benchmarks in Recovering Human Judgments of Similarity

Majority Choice



Note: The left panel depicts how often each measure recovers the human-selected more-similar speech. Our rhetorical cohesion measures matched human selections at higher rates than other text measures of related concepts. The 95% confidence intervals are generated via 1,000 hierarchical and paired bootstraps. The right panel depicts the pairwise accuracy differences with bootstrapped confidence intervals.

⁷We generate confidence intervals via hierarchical, paired bootstraps. We resample focal speeches, and comparison tasks within each sampled focal speech, with replacement and compute accuracy 1000 times.

4.2 Convergence with Feature-Specific Measures

Next, we assess whether our measure converges with the three conceptually related forms of textual alignment: expressed ideology, rhetorical partisanship, and lexical overlap. Each should be positively correlated with rhetorical cohesion. However, each of these measures captures one way speeches can align, so we expect these correlations to be modest.

Table 2: Correlations Among Rhetorical Cohesion Scores and Three Feature-Specific Measures

	OpenAI-3S	BGEM3-Dense	BGEM3-ColBERT	Expressed Ideology	Rhetorical Partisanship
OpenAI-3S	—				
BGEM3-Dense	0.891	—			
BGEM3-ColBERT	0.865	0.901	—		
Expressed Ideology	0.421	0.394	0.370	—	
Rhetorical Partisanship	0.221	0.214	0.225	0.100	—
Lexical Overlap	0.127	0.159	0.141	0.056	0.047

We present pairwise correlations in Table 2 and highlight three key results. First, all three embedding-based implementations of rhetorical cohesion appear in the upper-left panel of the table. They are highly correlated with one another, with coefficients ranging from 0.865 to 0.901. This result indicates our framework for measuring rhetorical cohesion is not especially sensitive to the selected embedding models.

Second, in the bottom left, we report correlations between rhetorical cohesion and three single-feature measures of text-alignment. These correlate positively, with modest to moderate magnitudes. Expressed ideology correlates at 0.395, rhetorical partisanship at 0.220, and lexical overlap at 0.142, on average. In Table A6, we regress all three versions of our measure on these features; the joint R^2 is 0.202 on average. Each feature-specific measure is positively correlated with rhetorical cohesion, but the individual correlations are generally modest, and the three do not jointly reproduce our measures.

Third, in the bottom-right, we observe that correlations between these three measures

are smaller than their correlations with rhetorical cohesion. These weak inter-correlations suggest that these measures capture different forms of alignment (as expected). Using any one as a proxy for rhetorical cohesion could lead a researcher to draw different substantive conclusions.

4.3 Known-Groups Validity: Southern Democrats

Throughout the mid-twentieth century, the Democratic party housed a distinct Southern faction whose ideological preferences diverged from the party's non-Southern faction (Key 1949). However, the Southern Realignment of the 1980s and 1990s led to the replacement of these conservative Democrats by conservative Republicans, increasing intra-party unity on both sides of the aisle (McCarty, Poole and Rosenthal 2008; Sinclair 2006). Given these patterns, we expect a valid measure of rhetorical cohesion to recover inter-factional differences between Southern and non-Southern Democrats. Same-faction pairs (i.e., two Southern Democrats or two non-Southern Democrats) should exhibit more rhetorical cohesion than a cross-faction pair (i.e., one Southern and one non-Southern Democrat) prior to the 104th Congress, what several scholars describe as a key turning point in congressional polarization (Lee 2016; Sinclair 2006; Theriault and Rohde 2011). These differences should attenuate after.

To test our hypothesis, we separate Democratic pairs into three groups: non-Southern Democratic pairs, Southern Democratic pairs, and cross-faction Democratic pairs.⁸ We regress the standardized ColBERT score on these pair indicators, an indicator for the post-103rd Congress period, and their interaction. Our models include topic Congress, and month fixed effects, and standard errors are lawmaker-dyad robust (Aronow, Samii and Assenova 2015). We present differences in predicted similarity between dyad types in Figure 4. The baseline group is a cross-factional pair. Higher values on the y -axis indicate that the specified same-faction pair exhibits higher rhetorical cohesion than a cross-faction

⁸Southern Democrats include House members from AL, AR, FL, GA, LA, MS, NC, SC, TN, TX, and VA.

pair.

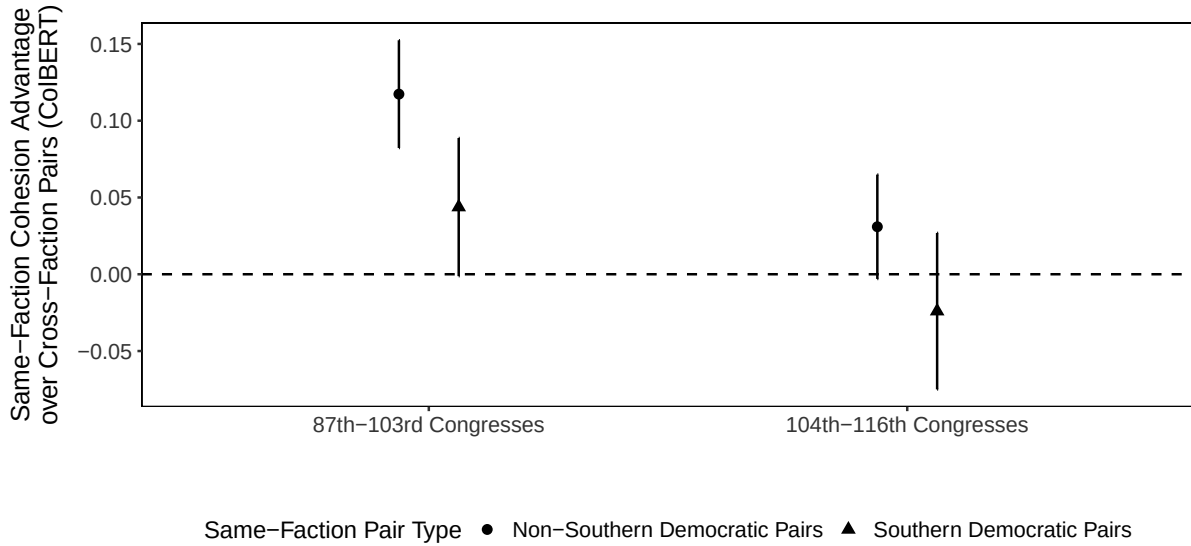


Figure 4: Two non-Southern Democrats exhibit higher rhetorical cohesion compared to a Southern/non-Southern Democratic pair before the 104th Congress. These differences in predicted similarity attenuate in the post-103rd Congress period.

On the left, we see that a non-Southern (Southern) Democratic pair is estimated to be 0.116 (0.045) standard deviations more rhetorically cohesive than a cross-faction pair before the 104th Congress. These differences are not statistically significant for Southern Democratic pairs, but non-Southern Democratic pairs are statistically more cohesive than a mixed-faction pair. In the 104th Congress and after, these differences attenuate and there are no statistical differences between Democratic pairings. These results provide evidence that our measure recovers expected factional differences and their subsequent attenuation.

5 Applications

To show how our measure can provide insight into congressional behavior, we consider four theory-grounded correlates of rhetorical cohesion in the 1994-2020 Congresses:

electoral proximity, majority status, issue salience, and seat safety.⁹ First, we examine how rhetorical cohesion varies with these political conditions. Then, we show how the conclusions change if we use a single-feature measure instead.

5.1 Correlates of Rhetorical Cohesion

Electoral incentives should increase rhetorical cohesion, especially when parties hold institutional power to set the legislative agenda. Party brands provide cheap information to voters about candidates' preferences and potential outcomes from party control of government (APSA 1950; Snyder and Ting 2002). Thus, co-partisans should be more interested in rhetorical cohesion as elections approach. Consistent with this argument, Weber and Parsons (2016) finds that roll-call cohesion increases as election day nears. Similarly, *we expect rhetorical cohesion to increase as congressional elections approach*. With an eye toward electoral success, parties that hold the majority have the ability to set the agenda, uniting their party and dividing the minority (Cox and McCubbins 1993, 2005; Sellers 2009). Therefore, *we expect higher rhetorical cohesion among majority-party members*.

The costs of defection should also vary across issues and constituencies. Parties should prioritize cohesive messaging on salient issues, which are most consequential for the brand (Groeling 2010; Sellers 2009). In a survey of Danish politicians, Tromborg (2019) finds that rank-and-file positions are closer to leaders' positions when issues are more salient. Similarly, *we expect rhetorical cohesion to be higher on more salient issues*. The party's message is likely to be a better fit in constituencies that are strongly co-partisan; marginal members face incentives to differentiate themselves (Canes-Wrone, Brady and Cogan 2002; Carson et al. 2010; Noble 2024). Therefore, *we expect safe-seat lawmaker-pairs to exhibit more rhetorical cohesion than mixed or marginal-seat pairs*.

To test these hypotheses, we regress rhetorical cohesion on variables representing

⁹This time period holds constant key features of the political environment, such as party polarization and electoral competition (Gentzkow, Shapiro and Taddy 2019; Lee 2016; Sinclair 2006).

these political conditions at the speech-dyad level as specified in Table 3. We also include topic, Congress, and month fixed effects and present models with and without lawmaker-dyad fixed effects. Standard errors are dyad-robust.

The results in Table 4 generally support our hypotheses. During the midterm election period, rhetorical cohesion increases by 0.048 standard deviations on average. However, these coefficients are small, near zero, and not statistically significant when using the BGEM-3 Dense dependent variable. Majority parties are also 0.053 standard deviations more rhetorically cohesive than minority parties on average. For issue salience, moving from the 25th percentile to the 75th percentile is associated with a 0.050 standard deviation increase in rhetorical cohesion on average. As expected, safe seat dyads are about 0.055 standard deviations more rhetorically cohesive than unsafe seat dyads, but only in models without lawmakers fixed effects. The effects of seat safety are cross-sectional, not driven by changing constituencies. Finally, the presidential election results are negative and sometimes statistically significant, at odds with our theoretical expectations. The marginal effects of elections in presidential years are close to 0. One possible explanation is that presidential campaigns complicate congressional messaging by forcing president-specific issues onto the agenda. Because we do not directly test this mechanism, we highlight this empirical pattern for future exploration. Ultimately the effect sizes and R^2 are modest, which makes sense given the range of factors that can affect speech similarity and the high-dimensionality of text data. Importantly, the signs and significance are consistent across specifications (with the aforementioned exceptions with respect to the electoral variables), indicating correlates are generally robust across embedding models. This consistency lends further credibility to our measure as results are not driven by the choice of a specific embedding model.

Table 3: Definition of Key Variables in Regression Models

Variable	Definition
<i>Political Conditions</i>	
Congressional Election Cycle	Continuous variable that is 0 on January 3rd of a new Congress and linearly increases to 1 on election day; excludes post-election period. (cf. Weber and Parsons 2016).
Pres. Election Congress	Indicator for Congresses that end in a presidential election year.
Majority Party	Indicator that the dyad’s party holds the House majority.
Issue Salience	Annual proportion of Gallup respondents identifying a problem in the corresponding CAP policy topic as the country’s most important problem (from Jones et al. 2023).
Safe Seat Dyad	Indicator that both lawmakers hold safe seats, defined as districts where their party’s presidential candidate received at least 65% of the vote in the previous election (following Carson et al. 2010).
<i>Controls</i>	
Republican	Indicator that both lawmakers are Republicans.
NOMINATE Similarity	Negative absolute difference between the lawmakers’ NOMINATE scores; higher values indicate more similar voting records.
Presidential Co-Partisans	Indicator that the lawmakers share the president’s party affiliation.
N. Leaders in Pair	Number of party leaders in the dyad, ranging from zero to two (from Volden and Wiseman 2014).
Same State	Indicator that both lawmakers represent the same state.
Speech-Length Difference	Absolute difference between the logged lengths of the paired speeches.

5.2 How the Choice of a Measure Shapes the Results

What would we conclude if we used an existing text-as-data measure as an alternative to our purpose-built measure of rhetorical cohesion? We answer this question in Figure 5. Here, we show relationships between our key independent variables and our cross-sectional ColBERT model (triangles, column 5 of the previous table) and three additional models using expressed ideology, rhetorical partisanship, and lexical overlap as alternative dependent variables, across a common sample with no missingness. The underlying data are exactly the same across models. The dependent variables are standardized. Full

Table 4: Correlates of Rhetorical Cohesion

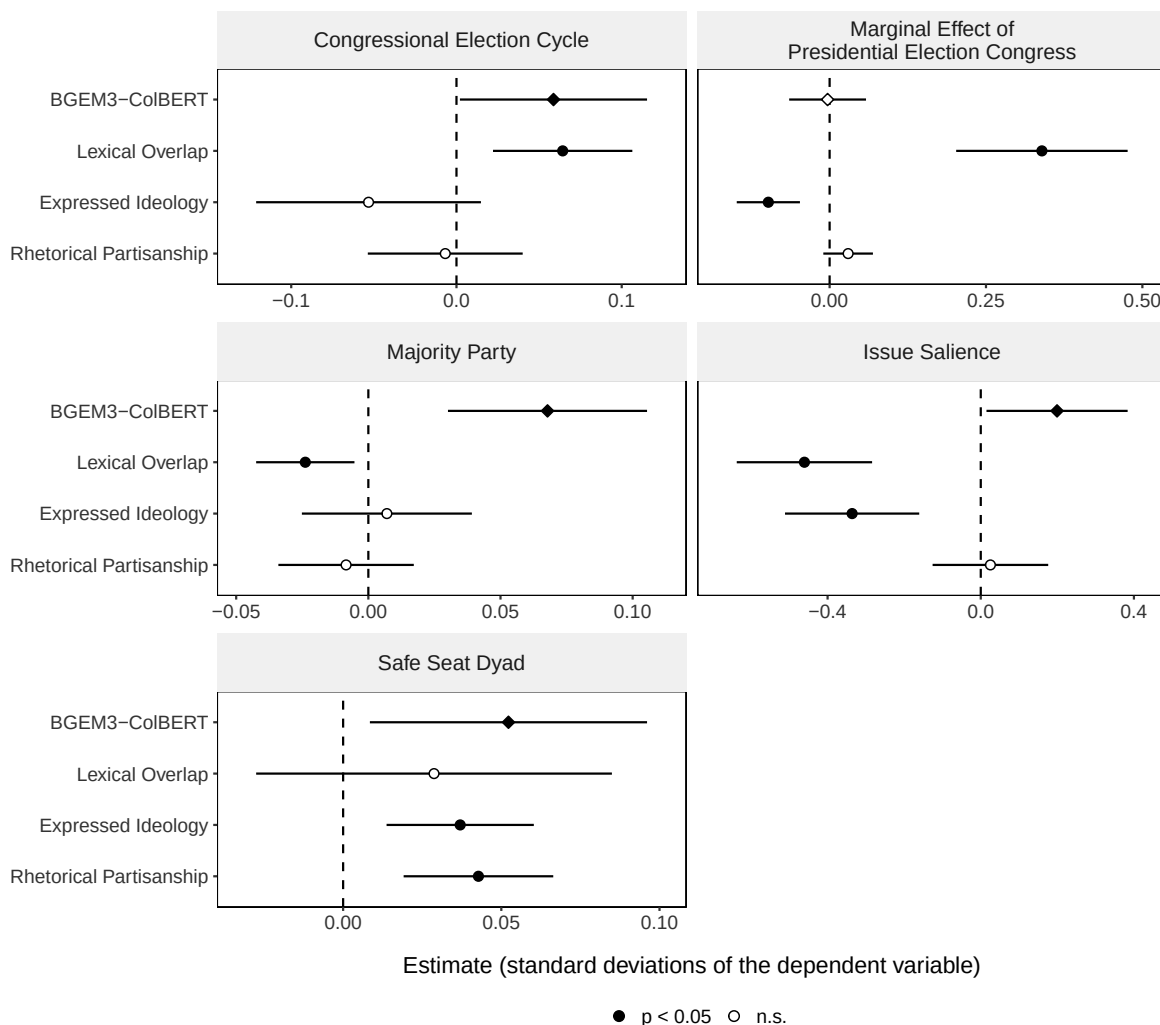
	OpenAI-3S		BGEM3-Dense		BGEM3-ColBERT	
	(1)	(2)	(3)	(4)	(5)	(6)
Congressional Election Cycle	0.089** (0.028)	0.076*** (0.021)	0.014 (0.027)	-0.001 (0.021)	0.059* (0.027)	0.049* (0.021)
Congressional Election Cycle x Presidential Election Congress	-0.123** (0.039)	-0.105*** (0.028)	-0.033 (0.038)	-0.013 (0.027)	-0.062 (0.040)	-0.048+ (0.029)
Majority Party	0.045** (0.017)	0.028* (0.013)	0.070*** (0.017)	0.057*** (0.013)	0.068*** (0.017)	0.050*** (0.013)
Issue Salience	0.418*** (0.090)	0.346*** (0.064)	0.365*** (0.091)	0.320*** (0.067)	0.199* (0.089)	0.143* (0.060)
Safe Seat Dyad	0.057** (0.021)	0.012 (0.012)	0.055** (0.020)	0.012 (0.014)	0.052* (0.022)	0.009 (0.014)
Republican	-0.169*** (0.028)		-0.180*** (0.027)		-0.189*** (0.028)	
NOMINATE Similarity	0.234*** (0.051)		0.247*** (0.049)		0.169** (0.052)	
Presidential Co-Partisans	0.003 (0.015)	-0.004 (0.011)	-0.003 (0.014)	-0.007 (0.011)	0.019 (0.015)	0.011 (0.011)
N. Leaders in Pair	0.108*** (0.023)	0.052* (0.022)	0.098*** (0.016)	0.043** (0.014)	0.101*** (0.023)	0.048** (0.018)
Same State	0.091*** (0.014)		0.094*** (0.014)		0.072*** (0.014)	
Length	-0.384*** (0.010)	-0.366*** (0.009)	-0.389*** (0.010)	-0.354*** (0.009)	-0.588*** (0.014)	-0.566*** (0.013)
Fixed Effects						
Lawmaker Dyad		✓		✓		✓
Topic	✓	✓	✓	✓	✓	✓
Congress	✓	✓	✓	✓	✓	✓
Month	✓	✓	✓	✓	✓	✓
Num.Obs.	1584520	1553534	1584520	1553534	1584520	1553534
R2 Adj.	0.141	0.255	0.149	0.254	0.220	0.323
R2 Within Adj.	0.091	0.079	0.089	0.071	0.178	0.170

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Results are from ordinary least squares models. Standard errors are lawmaker-dyad robust. The constitutive term for Presidential Election Congress is omitted as it is colinear with Congress fixed effects.

models can be found in Table B2.

Figure 5: Comparing Relationships Between Key Variables with Rhetorical Cohesion and Feature-Specific Measures of Textual Alignment



Note: Relationships between key independent variables and outcome measures vary when substituting feature-specific measures for rhetorical cohesion. Full models are in Table B2.

Comparing models, we can see that some of the relationships hold across specifications, while others differ. All measures show directionally similar correlations with seat safety, and to an extent, congressional midterms—although these correlations are not always statistically significant. Large differences can be seen, however, for majority party and issue saliency. Our measure is the only one to recover a positive effect of majority

control on rhetorical cohesion. However, lexical overlap is negatively associated with majority status and the other two coefficients are not significant. Similarly, where our measure is positively associated with issue salience, both lexical overlap and expressed ideology are negatively associated with issue salience, and rhetorical partisanship is null. As a consequence, substituting one of these measures for rhetorical cohesion can affect the substantive conclusions one draws.

These results are not intended to suggest that any feature-specific measure is misleading or inferior. Rather, we show that these four measures capture different ways in which speeches can align, and they are not substitutes. Researchers testing theories that concern similarity in partisan language, expressed political positions, or the specific reuse of phrases, should prefer those specific measures. Researchers studying overall rhetorical cohesion should use a measure validated for that purpose.

6 Conclusion

Rhetorical cohesion is an important but difficult-to-measure aspect of party communication. Party members can convey similar messages through shared positions, arguments, frames, or language, making any single observable textual feature inadequate to represent overall rhetorical similarity. We develop a two-stage framework using topic modeling and contextual embeddings to estimate rhetorical cohesion. We validate the resulting measures by showing that they closely match human judgments of speech similarity, correlate with existing feature-specific measures, and recover a well-documented historical pattern of factional differences. Our substantive analysis further shows that rhetorical cohesion varies systematically with theoretically motivated political conditions and that substituting feature-specific measures can lead to different conclusions.

The ability to measure rhetorical cohesion opens new avenues for studying party communication. Our substantive application demonstrates how the measure can be used

to study the strategic use of cohesive messaging by examining how rhetorical cohesion varies across political conditions. Beyond such variation, future research can investigate both the sources and consequences of rhetorical cohesion. On the source side, researchers could examine how increasingly studied dimensions of political communication—e.g., emotional appeals, partisan language, and moral rhetoric—contribute to cohesive messaging. On the consequence side, future work could ask whether rhetorical cohesion strengthens a party’s electoral brand and shapes voters’ perceptions. Although our application focuses on party cohesion in the U.S. Congress, the framework can be easily adapted to comparative legislative settings and measure rhetorical cohesion globally. With a consistent topic classification scheme—such as the Comparative Agendas Project, the resulting similarity measures could facilitate comparisons of rhetorical cohesion across countries and issues on a common scale.

This approach also has limitations. While our measure captures overall message similarity, it does not tell which features drive that similarity. Future work could pair our measure with LLM-assisted content analysis to identify the rhetorical dimensions associated with observed cohesion. Second, computing pairwise similarity among n documents requires $O(n^2)$ comparisons, which can be computationally costly for large corpora. This burden can be reduced through theoretically informed blocking on e.g., issues, time periods or groups.

In conclusion, our study demonstrates that while rhetorical cohesion is difficult to capture with any single observable feature of political text, it can be measured systematically. By combining customized topic modeling with contextual embeddings and multiple validation strategies, we show that the overall similarity of political messages can be recovered at scale without reducing rhetorical cohesion to any particular dimension. This framework enables researchers to measure rhetorical cohesion across issues, over time, and across political institutions. By extending the study of party cohesion from how members vote together to how they communicate together, our approach offers a

new way to understand when political actors speak with a common voice and why that voice matters.

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A Measurement of Rhetorical Cohesion

A.1 Keywords for keyATM Model

In Table A1, we present the set of topics and keywords used to fit our keyATM models. These keywords are the top 15 keywords ranked by tf-idf within the party platforms, treating each topic as a single document. The *parliamentary* and *other* topic keywords were generated by the authors.

Table A1: Topics and keywords used to fit the keyATM model

Category	Keywords
Agriculture	farm, rancher, farmer, ranch, export, commod, agricultur, fiber, grain, embargo, pariti, crop, livestock, food, wheat
Civil Rights	abort, disabl, gender, religi, sex, discrimin, religion, ballot, de-segreg, vote, equal, segreg, reproduct, marriag, racial
Culture	art, artist, endow, film, museum, danc, leisur, opera, orchestra, theatr, scholar, heritag, writer, scholarship, music, cultur
Defense	nato, nuclear, missil, weapon, ballist, veteran, iraq, soviet, treati, troop, korea, allianc, deploy, vietnam, arm
Domestic Commerce	antitrust, merger, mortgag, gambl, dodd, lend, patent, sba, theft, conglomer, ftc, frank, consum, small, loan
Education	student, classroom, teacher, math, tuition, parent, read, academ, graduat, teach, elementari, english, childhood, bilingu, secondari
Energy	oil, gas, coal, solar, energi, nuclear, electr, petroleum, atom, geotherm, opec, decontrol, wind, fossil, ethanol
Environment	speci, pollut, emiss, wetland, superfund, toxic, air, carbon, greenhous, esa, soil, brownfield, wildlif, fish, habitat
Foreign Trade	export, trade, tariff, currenc, negoti, textil, reciproc, monetari, bilater, nafta, china, agreement, protectionist, gatt, foreign
Government Operations	postal, district, columbia, lobbi, census, elector, mail, servant, statehood, ballot, incumb, branch, candid, vote, sunset, usp

Health	medicar, medicaid, patient, hiv, healthcar, drug, coverag, nurs, diabet, mental, cancer, medic, prescript, diseases, health
Housing	homeownership, slum, mortgag, fha, rent, rental, urban, tenant, homeless, rural, fanni, freddi, mac, mae, neighborhood
Immigration	immigr, refuge, undocu, deport, visa, alien, reunif, english, amnesti, newcom, flee, asylum, citizenship, illeg, admiss
International Affairs	israel, africa, soviet, taiwan, palestinian, east, arab, cuba, peac, korea, terrorist, ireland, asia, afghanistan, cuban
Labor	overtim, hartley, taft, pension, bargain, picket, employe, bacon, davi, collect, worker, arbitr, autom, osha, union
Law and Crime	gun, crime, crimin, drug, sentenc, offend, firearm, juvenil, polic, prison, victim, narcot, pornographi, traffick, marijuana
Macroeconomics	deficit, inflat, monetari, bracket, spend, debt, incom, wealthi, wealthiest, recess, taxat, loophol, inflationari, estat, code
Public Lands	puerto, indian, rico, guam, forest, nativ, hawaiian, tribal, virgin, samoa, mariana, tribe, miner, park, wilder
Social Welfare	welfar, parent, needi, nutrit, stamp, social, elder, child, recipi, disabl, lunch, older, charit, mother, poverti
Technology	space, nasa, broadband, internet, broadcast, scientif, telecommun, orbit, saturn, spacecraft, satellit, scienc, cyber, entertain, media
Transportation	highway, railroad, merchant, passeng, rail, freight, airport, transport, mode, maritim, congest, traffic, amtrak, marin, truck
Parliamentary	yield, gentleman, consent, amend, time, minut, senat, hous, bill, order, thank, committe, move, vote, quorum, motion, tabl
Other	peopl, go, get, got, laughter, know, thing, want, say, think, thank

A.2 Pre-Processing

To pre-process our speeches, we first subset to speeches with more than thirty words (following Noble 2024). We then tokenize to unigrams, remove numbers, punctuation, symbols, and separators, lowercase words, remove English stop words from the quanteda package and tokens with fewer than three characters, stem words, and remove words that appear fewer than 20 times or in fewer than 15 documents in each corpus. To match the corpus from Noble (2024), which we use for the 93rd–114th Congress, we exclude a small set of phrases from all Congresses. These include the president of the senate, the president pro tempore, ^ mr\. president, ^mister president, and ^madam[e]? president.

A.3 Topic Validation

To ensure our topic model and subsequent top-1 labels are accurate, we follow Ying, Montgomery and Stewart (2022) in conducting an Optimal Labeling (OL) task with two trained coders. One of the authors randomly sampled 105 speeches from the corpus (five per policy topic) and presented the coders (the other author and a trained research assistant, both blind to the model labeling) with each full speech and four potential labels. One of these was the top label assigned by the model, while the other three were selected from among the remaining possible topics.¹ Coders were asked “What is the main policy category for this speech?” and selected one of the options. The model matched the author and RA in 80% and 81% of cases respectively—accuracy similar to that of Ying, Montgomery and Stewart (2022). Inter-rater reliability between the two coders was substantial, achieving a Cohen’s- κ of 0.77. In Table A2, we provide additional summary statistics on our Optimal Label (OL) topic model validation task. In Table A3, we break down our OL accuracy by topic category.

¹We exclude the next two most probable topics given that several CAP topics overlap (e.g., macroeconomy and domestic commerce) and could cause confusion.

Table A2: Human Coder Validation Statistics

Statistic	RA	Author
Total Speeches Assigned	105	105
Correct Classifications	85	84
Overall Accuracy (%)	81	80
Inter-Rater Reliability		
Percent Agreement (%)	78.1	
Cohen's Kappa	0.77	
Kappa p-value	< 0.001	

Notes: Validation based on 105 manually coded speeches compared against the keyATM top-1 topic labels. Percent agreement calculated as proportion of speeches where both coders provided identical classifications.

Table A3: Category-Specific Accuracy by Coder

Topic Category	N	RA (%)	Author (%)
Agriculture	5	100	100
Civil Rights	5	100	100
Culture	5	100	80
Education	5	100	100
Environment	5	100	100
Health	5	100	100
International Affairs	5	100	80
Social Welfare	5	100	80
Defense	5	80	80
Domestic Commerce	5	80	80
Energy	5	80	100
Foreign Trade	5	80	60
Government Operations	5	80	80
Labor	5	80	60
Law & Crime	5	80	100
Macroeconomics	5	80	80
Technology	5	80	100
Housing	5	60	40
Transportation	5	60	60
Immigration	5	40	60
Public Lands	5	20	40
Overall Accuracy	105	81	80

Notes: Category-specific accuracy rates for each of the 21 topic categories. Each category contains exactly 5 speeches in the validation sample. Accuracy calculated as the percentage of speeches correctly classified within each category.

A.4 Construct Validity: Embedding-based Similarity Validation

A.4.1 Comparison Sampling

We sampled focal speeches and paired comparison texts across the 104-114th Congresses, subject to a few constraints. To validate within-focal speech similarity scores, not just across-focal speech similarity, we ensured that each focal speech appeared multiple times with different sets of comparison speeches. We also wanted to assess whether our human raters could discriminate between speech pairs that were more or less similar. To that end, we calculated the standard deviation of OpenAI-3S scores across our corpus and then determined, for each focal speech, how large was the difference of similarity scores between each of the comparison speeches. We grouped comparison tasks into buckets based on the size of this difference, such as below 0.25 of a standard deviation, 0.25–0.50 of a standard deviation, up to greater than 2 standard deviations. We sampled approximately 30 comparison tasks per bucket for a total of 249 tasks.

To conduct our validation exercise, we hired and trained an undergraduate RA to complete each task, as described below. During the summer of 2025, our RA completed 249 validation tasks. In the summer of 2026, we recruited crowd workers through Prolific.² We first invited a large pool of workers to complete ten pilot sets, each consisting of five comparison tasks. Based on their pilot performance, we selected four high-performing workers to complete the remaining tasks. Each task was independently coded by two crowd workers. To avoid using pilot responses that had contributed to worker selection, we excluded those responses and reassigned the corresponding tasks to workers who had not previously seen them. At the end of this process, each of the 249 tasks was coded by our undergraduate RA and two crowd workers. We define the human benchmark as the majority choice among these three raters.

A.4.2 Example Task

Below are the instructions we provided to the human raters in how to think about similarity when selecting speeches:

Your task is to help validate an automated measure of similarity between political speeches. Below, you'll be shown a series of sets of two **comparison speeches**. For each pair of comparison speeches, choose the one that you believe is **more similar**

²Prolific is a research platform for recruiting pre-screened participants, making it well suited for reliable human labeling of AI tasks.

to the **focal speech** above.

How to think about similarity:

Similarity in this context is **broad and multidimensional**. There is no single correct criterion. Instead, consider the following dimensions when making your judgment:

- **Topic:** Are the speeches about the same issue or subject?
- **Position:** Do the speakers take similar stances or make similar arguments?
- **Language:** Do they use similar words or phrases?
- **Tone/Sentiment:** Is the emotional tone or attitude similar (e.g., optimistic, angry, urgent)?
- **Style:** Do they share rhetorical features like repetition, emphasis, or structure?

You do **not** need to agree with the speaker or analyze the truth of the content—just assess overall similarity in meaning, purpose, and presentation.

There will often be some overlap across both comparison speeches. Use your best judgment to select the one that feels closer **overall** to the focal speech. There are no trick questions—your intuitive assessment is valuable for this task.

Please note: sometimes, you will encounter a repeated comparison speech paired with a different comparison speech. That is expected. Just keep choosing the most similar one.

Then, the human raters were presented with a series of tasks like the one below.

Primary Speech (1120072647): mr. chair. i rise today in strong opposition to the socalled "workforce democracy and fairness act" . the changes to union election procedures promoted in this bill are the exact opposite of the kind of fair and democratic policies that our working families need. instead of focusing on job creation and the revitalization of our middle class. the republicans in this chamber are once again promoting legislation that undermines the rights of american workers. this proposed legislation would limit the ability of the national labor relations board to interpret our nations labor laws and to protect workers right

to unionize. for over 75 years. the national labor relations act has guaranteed the rights of employees to organize and bargain collectively. or to refrain from such activity if they choose. during the new deal. our predecessors in this body created the national labor relations board as an independent agency charged with the oversight and enforcement of these rights. h.r. 3094. which overturns the rulings of the nlr. undermines its charge to maintain fair and democratic relationships between unions and employers. this legislation allows the problem of prolonged delays in union elections to continue unchecked by adding mandatory and arbitrary waiting periods. it seizes from workers the right to determine their own representative membership groups. which would allow unscrupulous businesses to suppress election drives and vote down union representation. it would also make it possible for irresponsible and frivolous litigation to endlessly delay the election process. effectively barring workers from their fundamental right to collective bargaining representation in the workplace. supporting and protecting americas workers is an essential part of rebuilding our economy and ensuring that all families and communities share in our nations prosperity. our middle class was built on the rights and safeguards that labor unions fought to obtain. from the 40 hour workweek to ending child labor. union representation has helped to guarantee rights that many of us take for granted today. unions negotiate for safe working conditions. living wages. and basic benefits that impact all workers. efforts to decrease the power of collective bargaining in this country in recent decades have been accompanied by an erosion of workers benefits and greater income inequality. this year in wisconsin and ohio. we have seen voters reject recent attempts to strip away the rights of government workers. and we should likewise reject this attempt to limit access to these rights for those in the private workforce. this bill does nothing to protect and support working families. and i urge my colleagues to stand up for workers rights and oppose this bill.

Comparison A (1120072704): i yield myself such time as i may consume. what i say to my good friend from south carolina is that i have the greatest respect for employers. id like the gentleman to join me in passing the american jobs act to give them payroll tax relief and to give them tax credits for hiring new employees. but you have to ask the question: after this bills implementation. will workers view their workplaces more favorably? will their wages match the growth rates of the companies and economy? will workers feel like american employers. supported by

government. provide meaningful safety for community survival? this legislation. frankly. undermines the american workers. can we all get along? can we find a way to address the concerns of making sure that we are fair to the employer but not have delay after delay after delay to deny someone his constitutional right of organizing freedom of expression? i think we can. the elimination of the provisions that i have spoken of is a dilatory upper hand of employers to get the better hand of our employees. i reserve the balance of my time. the acting chair. the gentlewoman from texas has 15 seconds remaining. and the gentleman from south carolina has 45 seconds remaining.

Comparsion B (1120072655): mr. chair. i rise today in opposition to h.r. 3094. the republican plan to crush workers rights and destroy any glimmer of hope our working families have at economic recovery. the republicans designed this bill to destroy 75 years of national labor review board case law in their attempt to dismantle the middle class. collective bargaining and the right to organize helped build a strong american middle class. it doesnt cost the federal government one dime in real money. instead of taking steps to create jobs and strengthen working families. republicans are dismantling key worker protections. all workers should have the ability to negotiate with their employer about salary and benefits. whether theyre in a union or not. organized labor is great for business. thousands of companies across the country thrive with a unionized workforce. those businesses recognize that their employees deserve to have a safe workplace and fair wages and benefits. thats just good business. this bill encourages corporations to stall nlr elections while they mount a onesided. antiunion campaign. at its core. this is an undemocratic bill that undermines our values. we have a long established process for workers to attempt to form a union and collectively bargain with employers. employers and employees should stay on equal ground in the process. there is no need to deny workers their right to a free and fair union election. many of my republican friends like to talk about the issue of tort reform. they like to tell us that we have to prevent frivolous lawsuits they cost taxpayers millions and millions of dollars and they drag down the economy. i have news for my republican friends: the election prevention act encourages frivolous litigation. this bill will mean mountains of litigation before union elections can be held. the result is a massive backlog. guess who picks up the tab? the american taxpayer! we have important issues facing our country and it boggles my mind that we are taking up yet another

bill that does nothing to get our friends and neighbors back to work. we need to focus on lowering the unemployment rate and creating jobsnot taking away the rights of hardworking americans. i urge my colleagues to recognize this veiled attempt to destroy the rights of american working families.

Which comparison speech (A or B) is more similar to the primary speech?

A: 1120072704

B: 1120072655

Here, our RA and the embedding model selected comparison B. As can be seen on a careful reading, comparison B is more similar to the focal speech for several reasons. First, although all speeches concern relationships between capital and labor, the focal speech and speech B explicitly discuss the NLRB whereas comparison A does not. Both the focal speech and comparison B also take a clear pro-labor position, whereas comparison A takes a more balanced approach favoring both employers and labor. Both the focal speech and comparison B are emotional, making use of hyperbole and extreme claims such as “irresponsible and frivolous litigation to endlessly delay” (focal speech) and “destroy any glimmer of hope our working families have” (comparison B). Comparison A makes use of a different rhetorical strategy, questioning. For these reasons, it is clear why both the RA and model selected comparison B.

A.4.3 Inter-coder Reliability

Across the 249 coding tasks, all three raters selected the same comparison speech in 63% of tasks. In Table A4, we present pairwise agreement statistics. Pairwise agreement ranged from 0.5 to 0.8 and Cohen’s κ ranged from 0 in one case to 0.6, with a median of 0.49, indicative of modest but heterogeneous inter-coder reliability. We attempted to keep worker pairs together to produce more precise pairwise reliability estimates. However, to complete the validation set without assigning workers tasks they had completed as part of the entry pilot, we created two smaller worker pairings. These smaller pairings showed lower agreement than the larger pairings.

As shown in Table A5, disagreement is concentrated among speeches that are closer to each other in the embedding space. For each embedding-based measure, we divide tasks into terciles based on the difference between the two focal-comparison speech similarity scores. Agreement increases across terciles: ranging from 39–43% in the lowest

Table A4: Pairwise Agreement Among Human Coders

Coder 1	Coder 2	Tasks	Agreement (%)	Cohen’s κ
Crowd Worker C	RA	125	80.0	0.597
Crowd Worker B	Crowd Worker C	96	79.2	0.581
Crowd Worker A	Crowd Worker D	104	76.9	0.538
Crowd Worker D	RA	133	75.2	0.507
Crowd Worker B	RA	116	74.1	0.483
Crowd Worker A	RA	124	71.8	0.436
Crowd Worker C	Crowd Worker D	29	69.0	0.368
Crowd Worker A	Crowd Worker B	20	50.0	0.000

tericle to 80–87% in the highest tercile. Thus, disagreement stems primarily from candidate speeches that are both relatively similar to the focal speech on our outcome measures rather than idiosyncrasies of our coders. We use majority vote of all three coders as our final human benchmark to both incorporate this disagreement and reduce the influence of any one coder’s particular interpretation.

Table A5: Share of Tasks with Unanimous Human Agreement (%)

Difference in Candidate Similarity Scores (Terciles)	OpenAI-3S	BGEM3-Dense	BGEM3-ColBERT
Lower Third	38.6	43.4	41.0
Middle Third	69.9	57.8	63.9
Upper Third	79.5	86.7	83.1

Note: Within each method, Lower Third, Middle Third, and Upper Third denote terciles of the difference between the two candidate similarity scores. Each cell is the percentage of the 83 tasks in that third on which the RA and both crowd workers selected the same comparison speech.

A.5 Joint Correlation Between Rhetorical Cohesion and Single-Feature Measures of Text Alignment

Table A6: Joint Correlation Between Rhetorical Cohesion and Single-Feature Measures of Text Alignment

	GPT (1)	Dense Dot (2)	ColBERT (3)
Partisan Language	0.025*** (0.000)	0.016*** (0.000)	0.017*** (0.000)
Lexical Overlap	0.014*** (0.000)	0.012*** (0.000)	0.011*** (0.000)
Topic-Wordfish Similarity	0.056*** (0.000)	0.035*** (0.000)	0.032*** (0.000)
Intercept	0.650*** (0.000)	0.692*** (0.000)	0.691*** (0.000)
Num.Obs.	1784116	1784116	1784116
R2 Adj.	0.219	0.203	0.185

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Results are from ordinary least squares models. All dependent variables are standardized.

B Additional Empirical Models

B.1 Southern Democrat Validation

Table B1: Democratic Faction Rhetorical Cohesion, 1961–2020

	(1)
Non-Southern Democrat Pair	0.116*** (0.023)
Southern Democrat Pair	0.045 (0.028)
Non-Southern Democrat Pair x Post-1994	−0.083** (0.032)
Southern Democrat Pair x Post-1994	−0.072+ (0.041)
Topic Fixed Effects	✓
Congress Fixed Effects	✓
Month Fixed Effects	✓
Num.Obs.	3031023
R2 Adj.	0.086
R2 Within Adj.	0.002

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Results are from ordinary least squares models. Standard errors dyad-robust.

B.2 Models Underlying Figure 5

Table B2: Political Correlates of Rhetorical Cohesion and Feature-Specific Measures of Textual Alignment

	BGEM3- Colbert	Expressed Ideology	Rhetorical Partisanship	Lexical Overlap
	(1)	(2)	(3)	(4)
Congressional Election Cycle	0.059* (0.029)	−0.053 (0.035)	−0.007 (0.024)	0.064** (0.022)
Congressional Election Cycle x Presidential Election Congress Majority Party	−0.062 (0.042)	−0.045 (0.044)	0.036 (0.031)	0.275*** (0.061)
Issue Salience	0.068*** (0.019)	0.007 (0.016)	−0.008 (0.013)	−0.024* (0.009)
Safe Seat Dyad	0.199* (0.094)	−0.336*** (0.089)	0.025 (0.077)	−0.461*** (0.090)
Republican	0.052* (0.022)	0.037** (0.012)	0.043*** (0.012)	0.029 (0.029)
NOMINATE Similarity	−0.189*** (0.031)	−0.062*** (0.018)	−0.014 (0.016)	−0.061*** (0.013)
Presidential Co-Partisans	0.169** (0.054)	0.194*** (0.033)	0.254*** (0.032)	0.051 (0.034)
N. Leaders in Pair	0.019 (0.016)	−0.023 (0.015)	−0.047*** (0.010)	0.071*** (0.014)
Same State	0.101*** (0.023)	0.054*** (0.012)	0.046*** (0.013)	0.025 (0.027)
Length	0.072*** (0.015)	0.040*** (0.009)	0.018+ (0.010)	0.052* (0.025)
	−0.588*** (0.015)	−0.110*** (0.007)	−0.102*** (0.006)	−0.109*** (0.018)
Fixed Effects				
Congress	✓	✓	✓	✓
Topic	✓	✓	✓	✓
Month	✓	✓	✓	✓
Num.Obs.	1584515	1584515	1584515	1584515
R2 Adj.	0.220	0.057	0.019	0.102
R2 Within Adj.	0.178	0.009	0.007	0.014

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Note: Results are from ordinary least squares models. Standard errors are lawmaker-dyad robust. We subset to complete cases across all four dependent variables.